

Julia Sisk-Reynes

Professional address

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Contact

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SAO/NASA ADS

Astronomer and postdoctoral researcher using multiwavelength data, strong gravitational lensing, and computational methods to investigate and characterize the incidence of dual active galactic nuclei in merging environments across the mass scale, the accretion flow onto supermassive black holes, and the nature of dark matter using current and future instrumentation.

EMPLOYMENT AND RESEARCH

- Sep 2026– **Postdoctoral Research Associate**
Physics Department, UMBC, 1000 Hilltop Circle, Baltimore, MD 21250
Supervisor: Prof. Adi Foord.
- Sep 2024–Aug 2026 **Postdoctoral Research Fellow**
Center for Astrophysics | Harvard & Smithsonian (CfA), Cambridge, MA
“Using gravitational lenses as high-resolution X-ray telescopes”. PI: Dr. Daniel Schwartz (NASA ADAP grant).
- Jun 2025– **Co-chair NewAthena Science Working Group 6 (SWG-6) on Cosmology & Fundamental Physics**
Chair: Dr. Esra Bulbul (MPE, Germany). Co-chair: Dr. Florian Pacaud (University of Bonn, Germany).

EDUCATION

- Oct 2020–Jul 2024 **Ph.D. in Astronomy**
Institute of Astronomy (IoA), University of Cambridge (UoC), UK
Thesis: “Unveiling fundamental physics with high-resolution X-ray spectroscopy of active galactic nuclei”.
Supervisors: Prof. Christopher Reynolds (University of Maryland College Park, MD) and Dr. James Matthews, (University of Oxford, UK).
- Sep 2016–Jun 2020 **MPhys (BSc + Master’s) in Physics** (First Class Honours)
The University of Manchester (UoM), UK
Master’s thesis: “Building models for the thermal Sunyaev-Zel’dovich effect from analytic and numerical cluster pressure profiles”.
Supervisor: Prof. Scott Kay (University of Manchester, UK).

AWARDS AND PRIZES

- Summer 2023 **Flatiron Institute Summer School on Fluid Dynamics**
Attended a two-week Fluid Dynamics Summer School at the CCA. Sponsored by the Simons Foundation.
- Jan 2022 **Paul Murdin Prize for the best published paper by a current student**
From IoA, University of Cambridge, UK, for [Sisk-Reynes et al., 2022, MNRAS, 510, 1](#)
- Oct 2020–Mar 2024 **Full Studentship from the UK Science and Technologies Facilities Council**
To pursue a doctorate in astronomy at the IoA, University of Cambridge, UK.
- Jun 2020 **Stellify Award from The University of Manchester, UK**
Honorary recognition for extracurricular contributions made as an undergraduate student 2016 – 2020.

AWARDED TELESCOPE TIME

- Aug 2026 **Science & cost PI**, *Chandra* General Observer Program, Cycle 28 – 160 ks.
Target: GraL J065904.1+16290 lensed quasar.
- Aug 2026 **Science & cost PI**, *Chandra* General Observer Program, Cycle 28 – 100 ks.
Target: DES J2321-463 lensed quasar.
- Jul 2024 **Co-PI**, *Chandra* General Observer Program, Cycle 26 – 120 ks.
Target: Zw2089 cluster. PI: Dr. Helen Russell, University of Nottingham, UK.
- Jul 2024 **Co-I**, *Chandra* Legacy Program, Cycle 26 – 3 Ms.
Target: Perseus Cluster. PI: Prof. Andrew Fabian, IoA, University of Cambridge, UK.

TECHNICAL SKILLS

Programming	Bash, Python, tel, Unix/Linux
Astronomy software	(Py)XSPEC, Sherpa, CIAO, ds9, SAOTrace (<i>Chandra</i> ray-trace simulations)
Numerical tools	ALPRO (axion code, advanced level), AREPO (basic level)
Languages	English, Catalan, Spanish

PRESENTATIONS

May 2022–	Fourty-two presentations at international conferences, workshops, telescope collaborations, and group meetings. Selected recent invited presentations:	
Aug 2026	Talk on behalf of <i>NewAthena</i> SWG6, E1.18, 46 th COSPAR Scientific Assembly	<i>Florence, Italy</i>
May 2026	Talk on lensed quasars, MIT Monday Afternoon Talks, MKI	<i>Cambridge, MA</i>
Apr 2026	Colloquium-style talk on lensed quasars, CfA UMBRELA Dialogues	<i>Cambridge, MA</i>
Apr 2026	Talk on a survey of supermassive black hole spins, BHEX spin demographics SWG	<i>Virtual</i>
Dec 2025	Colloquium-style talk, Lin Group Meeting, Hong Kong University	<i>Virtual</i>
Oct 2025	Colloquium, Universidad Nacional Autónoma de México, Mexico	<i>Virtual</i>

ACADEMIC, RESEARCH, AND RELATED ACTIVITIES

Scientific membership

May 2025–	Co-chair, <i>NewAthena</i> SWG6 - Cosmology and Fundamental Physics
May 2025–	Member, <i>NewAthena</i> SWG2 - Galaxies and Supermassive Black Holes
Dec 2024–	Member, <i>AXIS</i> Working Group on Active Galactic Nuclei
May 2023–Oct 2024	Member, <i>STROBE-X</i> Working Group on Cosmology

Peer Review

Aug 2025	Telescope Allocation Time - <i>XRISM</i> Guest Observer program reviewer, Cycle 2 (3 days)	<i>Baltimore, MD</i>
Mar–May 2025	Telescope Allocation Time - <i>Chandra</i> Guest Observer program reviewer, Cycle 27 (4 days)	<i>Virtual</i>
Nov 2024	Telescope Allocation Time - <i>NICER</i> Guest Observer program reviewer, Cycle 7 (5 days)	<i>Virtual</i>
Sept 2025–	Journal paper reviewer - Physical Review D, Physical Review Letters, MNRAS	

Press Coverage

May 2023	<i>Athena</i> Press Release on Sisk-Reynes et al., 2023, <i>ApJ</i> , 930, 1
Jun 2022	<i>Chandra</i> Press Release on Sisk-Reynes et al., 2022, <i>MNRAS</i> , 514, 2

Training

Sep 2025	Black Hole Spins Workshop, Wake Forest University (2 days)	<i>Winston-Salem, NC</i>
Feb 2025	3rd <i>XRISM</i> Community Workshop (3 days)	<i>College Park, MD</i>
Jul–Aug 2019	Summer research project on direct dark matter searches with <i>DarkSide-20k</i>	<i>UoM, UK</i>

Recent invited outreach presentations

Jul 2025	Speaker, Spanish Public Observatory (70 attendees)	<i>CfA, Cambridge, MA</i>
Feb 2025	Speaker on visit to a bilingual school (100 attendees)	<i>Chelsea, MA</i>
Mar 2024	Panelist, International Women's Day Discussion Panel, Women in Science	<i>IoA, UoC, UK</i>
Mar 2024–2023	Speaker, Lindemann Trust Science Day (120 attendees)	<i>Cambridge, UK</i>

Teaching and student supervision

Oct 2022–Jun 2023	Teaching - Mathematics I Supervisor, Emmanuel College	<i>UoC, UK</i>
Jan–May 2021	Teaching - Astrophysical Fluid Dynamics. Trinity College	<i>UoC, UK</i>
Sep 2017–May 2019	Student Support - Peer Assisted Study Sessions (PASS) Leader	<i>UoM, UK</i>
Sep 2025–Jun 2026	Student supervision - Main supervisor to Henry Hiscocks, exchange master's student	<i>CfA, Cambridge, MA</i>
Sep 2025–Jun 2026	Student supervision - Co-supervisor to Charlotte Eades, exchange master's student	<i>CfA, Cambridge, MA</i>
Nov 2024–Jun 2026	Student supervision - Advisor to Alex Ostridge, exchange master's student	<i>CfA, Cambridge, MA</i>
Jun–Aug 2022	Student supervision - Advisor to Esin Gulhabar, summer research student	<i>IoA, UoC, UK</i>

Selected institutional service

Apr 2025–	Lead organizer, monthly discussions, cosmology journal club	<i>CfA, Cambridge, MA</i>
Nov 2024–	Organizer, weekly HEAD seminar series	<i>CfA, Cambridge, MA</i>
Jul 2024	Local Organizing Committee, IoA 50-year anniversary conference	<i>IoA, UoC, UK</i>
Oct 2021–Apr 2024	Lead organizer, IoA International Women's Day (attendees: 100/year)	<i>IoA, UoC, UK</i>
Oct 2021–Apr 2024	Lead organizer, Newnham College MCR Speaker Seminar Series	<i>UoC, UK</i>
Mar 2017–Jun 2020	Lead committee member, undergraduate Nuclear & Particle Physics Society	<i>UoM, UK</i>